

TenderHub Ingestion Adapters Guide

Architecture, Specifications, Capabilities & Operations Manual

VERSION 2.4

PRODUCTION GRADE

ARCHITECTURE	TOTAL ADAPTERS	ENGINE STACK	CLOUD PERSISTENCE
Hexagonal / Adapter	5 Specialized Classes	Playwright + Ghostscript	Atlas + Cloudflare R2

1. Executive Architecture & Ingestion Philosophy

TenderHub implements a decoupled **Adapter Architecture** to communicate with the Jammu & Kashmir e-Procurement Portal (`jktenders.gov.in`). Crawling state-level government infrastructure requires handling multi-department hierarchies, nested NIT work packages (sibling tenders), variable captcha formats, asynchronous document opening windows, and large tender dossiers (> 5MB).

To maximize reliability, throughput, and minimize server overhead, the ingestion engine enforces a **strict separation of roles**:

Core Architectural Rule: `JKTenderAdapter` is used *strictly and exclusively* when crawling the entire catalog of active tenders across all 31 government organisations. In all other operational workflows (daily incremental syncs, scheduled cron cycles, targeted date backfills, and missing PDF document recovery), the system *always* utilizes `JKTenderDataAdapter`.

2. Complete Adapter Landscape & Capability Matrix

ADAPTER CLASS	PRIMARY ROLE & TARGET USE CASE	PORTAL TARGET	DOCS & BOQ	R2 ARCHIVE	EXECUTION MODEL
<code>JKTenderAdapter</code>	Bulk Active Crawler: Scrapes all active tenders across all 31 organisations. Handles multi-tender sub-tables.	<code>FrontEndTendersByOrganisation</code>	YES (Ghostscript >5MB)	YES (Docs + <code>tender.json</code>)	Heavy Bulk Script (Manual / Quarterly)
<code>JKTenderDataAdapter</code>	Targeted Date & Recovery Engine: Ingests tenders for a specific published date; recovers pending docs.	<code>FrontEndTendersByOrganisation</code> (Date Filtered)	YES (Ghostscript >5MB)	YES (Docs + <code>tender.json</code>)	Automated Crons (5x Daily) & Worker
<code>JKTenderMetadataAdapter</code>	Metadata Enrichment Engine: Fast table & date extraction without downloading binaries or solving PDF captchas.	Direct Tender Detail Pages	NO (Preserves R2 links)	YES (Fresh <code>tender.json</code>)	Maintenance & Date Repair Utility
<code>DummyTenderAdapter</code>	Sandbox Mock Adapter: Generates realistic synthetic tenders with simulated network delays for offline testing.	Synthetic In-Memory	NO (Mock URLs)	NO (Offline)	CI/CD Pipelines & Frontend Development
<code>TenderSourceAdapter</code>	Abstract Interface: Enforces contract compliance (<code>fetchList</code> , <code>fetchDetail</code> , <code>normalize</code>).	Abstract Contract	N/A	N/A	Base Class Inheritance Blueprint

3. End-to-End Ingestion Pipeline Architecture

1. Portal Crawl & Captcha Resolution	2. Document Download & Compression
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Headless Playwright Chromium navigates portal tables. Live captchas are solved using a triple-redundancy pipeline: **Google Gemini Flash Vision** → **Hugging Face Router Vision** → **Local Tesseract OCR**.

PDFs, ZIPs, and XLS BOQ files are streamed into memory. PDFs exceeding 5MB are automatically compressed via **Ghostscript (150 DPI)**, preventing network bloat while preserving clarity.

3. Atomic MongoDB Atlas Upsert

Tenders are persisted via `findOneAndUpdate`. Subdocument arrays (`nitDocuments`, `workItemDocuments`) are merged without duplicates. Sibling packages are linked via `baseTenderId`.

4. Cloudflare R2 Hierarchical Archiving

Documents and self-describing `tender.json` bundles are mirrored into deterministic S3/R2 keys:
`tenders/{deptCode}/{tenderId}_{publishedDate}/{file}` for zero-dependency disaster recovery.

Core Crawling Engines: Detailed Specifications

JKTenderAdapter (Organisation Active) vs. JKTenderDataAdapter (Date & Recovery)

CORE CRAWLERS

PLAYWRIGHT V1.62+

1. JKTenderAdapter — Bulk Active Organisation Crawler

File: `backend/src/services/adapters/JKTenderAdapter.js` (~103 KB)

Role: Full-breadth catalog crawler designed to traverse all 31 government organisations listed on FrontEndTendersByOrganisation, discover every active tender, download all associated documents, compress oversized files, and save the data to MongoDB Atlas and Cloudflare R2.

Key Capabilities & Hardening

- **Intermediate Multi-Tender Sub-Table Traversal:** Detects when a clicked tender expands into an intermediate "Tender List : Open Tender" sub-table. Iterates every individual work package, extracts metadata, downloads documents, establishes sibling linkages, and executes `navigateBackFromTenderDetails(page)` to ensure complete package processing.
- **100% Critical Date Accuracy & Reference Clamping:** Extracts Critical Dates directly from innermost table nodes. Clamps `publishedDate` to not exceed document download start or bid submission start dates, guarding against session captcha clock leak.
- **Mandatory Bidder Checklist:** Extracts `otherImportantDocuments` (Standard Bidder Certificate checklist tables).
- **Ghostscript Compression:** Automatically intercepts downloaded PDFs >5MB and runs `compressPDF()` via Ghostscript, shrinking files by up to 75% before R2 upload.
- **Self-Describing `tender.json`:** Directly executes `saveTenderAndUploadR2(item, hasDocuments, docReason)` to persist an atomic upsert in MongoDB Atlas and archive an immutable JSON manifest into Cloudflare R2.

Execution Commands & NPM Scripts

```
# 1. Full portal crawl across all 31 active organisations (Default production run)
$ npm run scrape:all
# Underlying command: node src/scripts/fetchAllActiveTenders.js

# 2. Quick organization scrape with custom department filtering
$ npm run scrape:tenders -- --dept="PWD" --limit=20
# Underlying command: node src/scripts/fetchTenders.js

# 3. Interactive debugging run with visible browser window
$ node src/scripts/fetchAllActiveTenders.js --headless=false --limit=5
```

2. JKTenderDataAdapter — Targeted Date & Recovery Engine

File: `backend/src/services/adapters/JKTenderDataAdapter.js` (~86 KB)

Role: The daily workhorse of TenderHub. Designed for pinpoint date-based ingestion, scheduled cron jobs, and post-opening document recovery.

Key Capabilities & Hardening

- **Targeted Date Filtering:** Accepts `targetDate`: 'today' or explicit dates like '23-Sep-2026'. Filters organization rows at the portal list level, skipping historical records and reducing crawl times from hours to minutes.
- **Automated Document Recovery (`fetchPendingDocuments`):** Tenders published prior to their document download start date are recorded in `PendingDocumentTender`. This adapter scans that queue, visits opening tenders once their window unlocks, retrieves all PDFs/BOQs, and updates MongoDB/R2.
- **Worker Queue Integration:** Powers the BullMQ background worker (`tenderSync.worker.js`) across 5 automated daily ingestion windows: **9:00 AM, 10:00 AM, 1:00 PM, 3:00 PM, 6:30 PM IST**, plus hourly document retry sweeps.
- **Deterministic Storage Keys:** Passes explicit `deptCode` to guarantee canonical R2 storage paths: `tenders/{deptCode}/{tenderId}_{publishedDate}/{file}`.

Execution Commands & NPM Scripts

```
# 1. Ingest all active tenders published today (Daily production run)
$ npm run scrape:latest
# Underlying command: node src/scripts/fetchLatestDailyTenders.js

# 2. Ingest tenders for a specific calendar date (Backfill / Historical catchup)
$ npm run scrape:date -- --date="22-Sep-2026" --limit=100
# Underlying command: node src/scripts/fetchTendersByDate.js

# 3. Recover missing documents for tenders whose download windows have opened
$ npm run fetch:pending-docs
```

```
# Underlying command: node src/scripts/fetchPendingDocTenders.js
```

```
# 4. Start the BullMQ background sync worker (Runs all scheduled crawls)
```

```
$ npm run worker
```

Specialized Adapters & Architectural Contracts

JKTenderMetadataAdapter, DummyTenderAdapter & TenderSourceAdapter Base Class

UTILITY ENGINES

SCHEMA INTEGRITY

1. JKTenderMetadataAdapter — High-Speed Metadata Engine

File: backend/src/services/adapters/JKTenderMetadataAdapter.js (~28 KB)

Role: High-throughput, non-intrusive metadata synchronization engine. Used when schema fields, tender status, critical dates, or authority details must be refreshed across hundreds of existing database records without incurring the overhead of downloading multi-megabyte PDFs or solving download captchas.

Key Capabilities & Performance Benefits

- Binary Download Bypass:** Completely skips PDF/ZIP/XLS document downloading. By avoiding file transfers and download captchas, it processes tenders at **10x to 15x higher velocity** than full crawlers.
- R2 Link Preservation:** Strictly preserves all existing Cloudflare R2 links (pdfURLs, nitDocuments, workItemDocuments) stored in MongoDB, updating only textual metadata, tender value, closing dates, and authority contacts.
- Cloudflare R2 JSON Refresh:** Packages updated metadata into a fresh tender.json and uploads it directly to the tender's Cloudflare R2 archive, keeping Cloudflare synchronized with MongoDB Atlas.
- Guaranteed Date Extraction:** Specifically engineered to prevent session clock leaks by parsing inner date tables and rejecting top-level header timestamp fallbacks.

Execution Commands & NPM Scripts

```
# 1. Enrich existing MongoDB tenders with updated metadata & fresh tender.json in R2
$ npm run enrich:metadata
# Underlying command: node src/scripts/enrichExistingTenders.js

# 2. Enrich tenders with a specific batch limit
$ node src/scripts/enrichExistingTenders.js --limit=50

# 3. Synchronize all portal metadata directly without modifying documents
$ node src/scripts/syncAllMetadata.js
```

2. DummyTenderAdapter — Mock Sandbox & Testing Adapter

File: backend/src/services/adapters/DummyTenderAdapter.js (~2.5 KB)

Role: Sandbox adapter that produces synthetic tender structures matching the exact Mongoose schema.

Capabilities & Use Cases

- Offline Development:** Enables full frontend UI, backend API, and search filtering development without requiring internet connectivity or risking portal IP blocks.
- Network Latency Simulation:** Injects a configurable 1.5-second async delay to emulate realistic portal latency for testing frontend loading spinners, skeletons, and error boundaries.
- CI/CD Test Suites:** Used in automated unit testing pipelines where live portal scraping is prohibited.

Code Usage Example

```
// Usage in test suites or local seed scripts
import { DummyTenderAdapter } from './src/services/adapters/DummyTenderAdapter.js';
const adapter = new DummyTenderAdapter();
const mockTenders = await adapter.fetchList(1);
const normalized = adapter.normalize(mockTenders[0]);
```

3. TenderSourceAdapter — Abstract Base Contract

File: backend/src/services/adapters/TenderSourceAdapter.js (~0.6 KB)

Role: The abstract interface that enforces uniform contracts across all current and future tender portals (e.g., GeM, Central CPPP, Railway e-Procurement).

Contract Methods

METHOD SIGNATURE	CONTRACT REQUIREMENT	RETURN TYPE
constructor(portalName)	Initializes portal identifier; prevents direct instantiation of abstract class.	void
fetchList(page, filters)	Navigates catalog tables and returns raw tender listings for a page.	Promise<Array>

METHOD SIGNATURE	CONTRACT REQUIREMENT	RETURN TYPE
fetchDetail(sourceTenderId)	Navigates to detail page, solves captchas, extracts full metadata and documents.	Promise<Object>
normalize(rawTenderData)	Transforms portal-specific DOM keys into standardized TenderHub Mongoose schema.	Object (Schema Compliant)

Operational Playbook & CLI Cheat-Sheet

Production Workflows, Scenario Guides & Environment Reference

RUNBOOK READY

NODE.JS V22+

1. Scenario-Based Command Cheat-Sheet

OPERATIONAL SCENARIO	RECOMMENDED COMMAND	ACTIVE ADAPTER	EXPECTED OUTCOME
Daily Automated Sync	npm run scrape:latest	JKTenderDataAdapter	Ingests all active tenders published today across all 31 orgs.
Complete Portal Refresh	npm run scrape:all	JKTenderAdapter	Comprehensive bulk crawl of all active tenders and multi-work packages.
Date Range Backfill	npm run scrape:date -- --date="DD-MMM-YYYY"	JKTenderDataAdapter	Ingests tenders for a specific past date (e.g. 20-Sep-2026).
Unlock Pending Documents	npm run fetch:pending-docs	JKTenderDataAdapter	Backfills PDFs/BOQs for tenders whose download window opened today.
Metadata Schema Refresh	npm run enrich:metadata	JKTenderMetadataAdapter	Refreshes critical dates & metadata; updates tender.json in R2.
Continuous Daemon Worker	npm run worker	JKTenderDataAdapter	Background BullMQ runner executing 5 daily crawl slots + retry crons.
Inspect Sibling Tenders	node src/scripts/listMultiTenders.js	MongoDB Aggregation	Lists all multi-tender groups sharing base NIT IDs in the database.
Adapter Parity Regression	node tests/adapter_parity.test.js	Integrity Test	Verifies method parity, R2 key generation, and schema normalization.

2. Disaster Recovery & Manual Generation Commands

```
# Generate all executive documentation manuals (Deployment, Disaster Recovery, Crawl & Adapters)
$ npm run manuals:all

# Individual PDF manual generators
$ npm run manual:adapters # Generates this Architecture & Operations Guide PDF
$ npm run manual:disaster # Generates Disaster Recovery & Failover Manual PDF
$ npm run manual:deployment # Generates Production Deployment Playbook PDF
$ npm run report:pdf # Generates Executive Crawl Report PDF
```

3. Mandatory Environment Variables Configuration

VARIABLE KEY	REQUIRED BY	DESCRIPTION & EXAMPLE
MONGO_URI	All Adapters	Primary MongoDB Atlas connection string (Cluster0).
REDIS_URL	Worker & Queue	Upstash / local Redis URL for BullMQ job queue management.
R2_ACCESS_KEY_ID / R2_SECRET_ACCESS_KEY	All Adapters	Cloudflare R2 API credentials for document and JSON uploads.
R2_BUCKET_NAME	All Adapters	Target R2 bucket name (e.g. tenderhub-production).
GEMINI_API_KEY / GEMINI_MODEL	Captcha Service	Google Gemini Flash Vision API key (primary captcha solver).
HF_TOKEN / HF_MODEL	Captcha Service	Hugging Face Router Vision token (secondary captcha solver fallback).
SCRAPER_HEADLESS	All Adapters	Set to false for visual browser debugging; defaults to true.

System Health Verification: All adapters run under Node.js v22+ and require Playwright Chromium binaries. Ghostscript (gswin64c.exe or gs) must be installed on the system path for PDF compression (>5MB).

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